

MedBERT: A Pre-trained Language Model for Biomedical Named Entity Recognition

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Abstract

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Abstract:

This paper introduces MedBERT, a new pre-trained transformer-based model for biomedical named entity recognition. MedBERT is trained with 57.46M tokens collected from biomedical-related data sources, i.e. datasets acquired from N2C2, BioNLP, CRAFT challenges, and biomedical-related articles crawled from Wikipedia. We validate the effectiveness of MedBERT by comparing it with four publicly available pre-trained models on ten biomedical datasets from BioNLP and CRAFT shared tasks. Our experimental results show that models fine-tuned on MedBERT achieve state-of-the-art performance in nine datasets that predict Protein, Gene, Chemical, Cellular/Component, Gene Ontology, and Taxonomy entities. Specifically, the model achieved an average of 84.04% F1-micro score on ten test sets from BioNLP and CRAFT challenges with an improvement of 3.7% and 7.83% as compared to models that were fine-tuned on BioBERT and Bio_ClinicalBERT, respectively.

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